

AUTOMATED WASTE MANAGER : SMART DUSTBIN

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CERTIFICATE

This is to certify that the Mini Project entitled “Automated Waste Manager : Smart Dustbin” is a bonafide work of student Devesh Shelatkar-2304055, Priyanka Vaidya-2304063, Suyash Shirsat-2304056 submitted in the department of Electronics and Telecommunication Engineering in partial fulfilment of the requirement of academic curriculum of Semester -IV Mini-Project-1B as per the revised syllabus 2019C scheme.

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PROJECT REPORT APPROVAL

This project report entitled “Automated Waste Manager : Smart Dustbin” is approved for partial completion of Mini Project-1B in Sem -IV in Electronics & Telecommunication Engineering.

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ABSTRACT

The "Automated Waste Manager: Smart Dustbin" project presents an innovative approach to modernizing waste disposal through automation and sensor integration. By employing an ultrasonic sensor interfaced with a microcontroller (such as NodeMCU), the system detects the presence of waste or a user's hand, triggering a servo motor to automatically open the dustbin lid. This touchless operation enhances hygiene by minimizing physical contact, thereby reducing the risk of contamination. The smart dustbin is designed to be cost-effective and energy-efficient, making it suitable for deployment in various settings, including homes, offices, hospitals, and public spaces. By streamlining the waste disposal process and promoting sanitary conditions, this project contributes to improved public health and environmental cleanliness.

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1.INTRODUCTION

Overview:

Automated Waste Manager : Smart Dustbin

The Automated Waste Manager: Smart Dustbin is an IoT-based system designed to modernize waste disposal through automation and intelligent sensing. It combines real-time waste level monitoring with automatic lid control using obstacle detection, significantly improving hygiene and operational efficiency. By integrating sensors, a microcontroller, and platforms like Blynk and ThingSpeak, the system enables touchless operation, remote monitoring, and real-time data visualization.

The system consists of:

- Ultrasonic Sensor (Top) – Detects nearby obstacles (like a hand) to automatically open/close the lid
- Ultrasonic Sensor (Inside) – Measures the waste level to prevent overflow
- Microcontroller (NodeMCU) – Processes sensor data and controls the motor and communication
- Motor (Servo/DC Motor) – Operates the lid mechanism
- Blynk App – Provides real-time control and notifications via smartphone
- ThingSpeak Platform – Collects and visualizes sensor data over time for analysis and insights.

2. LITERATURE SURVEY

Problem Statement:

With the rise of smart cities and sustainability goals, traditional waste disposal systems face serious limitations. Manual monitoring of dustbins often results in inefficiencies—either bins overflow due to delayed collection or resources are wasted through premature pickups. This leads to hygiene concerns, environmental pollution, and higher operational costs.

The Automated Waste Manager: Smart Dustbin addresses these issues by integrating automation and IoT technologies into a single solution. It uses ultrasonic sensors to detect both nearby obstacles (for automatic lid opening) and internal waste levels (to prevent overflow). A NodeMCU microcontroller processes this data, while the Blynk App and ThingSpeak platform enable real-time monitoring and remote alerts.

Unlike commercial smart bins, which are often expensive and complex, this system is cost-effective, easy to maintain, and scalable. It provides valuable data insights that help optimize collection schedules, reduce fuel and labor costs, and improve hygiene through touchless operation—especially important in a post-pandemic world.

Additionally, this project has strong educational value. It offers a hands-on learning experience in IoT, automation, and smart infrastructure, making it ideal for students, DIY enthusiasts, and community-driven initiatives.

By combining affordability, functionality, and smart features, the Automated Waste Manager serves as a practical and innovative step toward efficient, data-driven waste management.

PROBLEM SOLUTION FOR MINI PROJECT:

LITERATURE REVIEW:

The Automated Waste Manager: Smart Dustbin provides a comprehensive, tech-driven solution to the inefficiencies of traditional waste disposal systems in the context of smart cities and sustainability goals.

By leveraging IoT and automation, the system integrates the following components:

- Ultrasonic Sensors to:
 - Detect waste levels inside the bin and prevent overflow.
 - Sense nearby motion to trigger automatic lid opening, ensuring a hygienic and touchless user experience.
- NodeMCU Microcontroller acts as the brain of the system, collecting data from sensors and transmitting it to cloud platforms.
- Blynk App & ThingSpeak Integration for:
 - Real-time monitoring of dustbin fill levels.
 - Remote alerts for timely waste collection, reducing manual effort and missed pickups.

This cost-effective system replaces outdated manual monitoring with smart, real-time solutions. It is:

- Scalable, making it adaptable to various urban and rural settings.
- Affordable, unlike many commercial alternatives.
- User-friendly and educational, offering a valuable learning platform in the fields of IoT, automation, and sustainability.

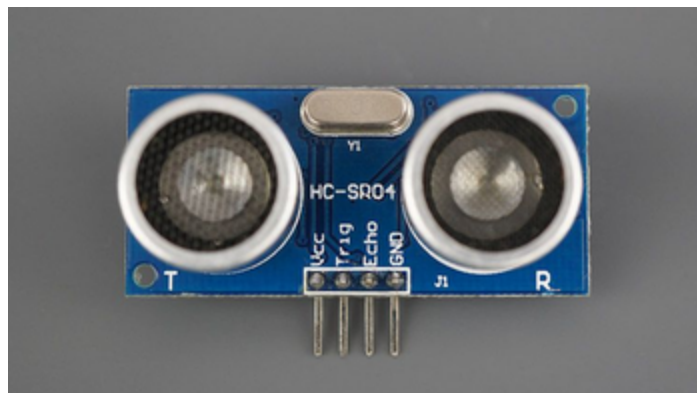
Through efficient data collection and processing, the system helps optimize waste collection schedules, minimize operational costs, and improve public hygiene, contributing to the development of cleaner, smarter cities.

HARDWARE USED:

1. Ultrasonic Sensors (×2)

Ultrasonic sensors are non-contact distance measuring devices that work by emitting ultrasonic waves and measuring the time it takes for the echo to return after bouncing off an object. In the Automated Waste Manager, two ultrasonic sensors are employed for different purposes:

- **Sensor 1 – Motion Detection (Top of Bin):**
 - This sensor is mounted near the upper edge or lid of the bin. It continuously scans for the presence of a person or object approaching the bin. When motion is detected within a specified range (e.g., 20–30 cm), it triggers the opening of the lid via the servo motor. This mechanism provides a contactless, hygienic user experience, particularly beneficial in public areas to reduce the spread of germs and viruses.
- **Sensor 2 – Fill Level Monitoring (Inside the Bin):**
 - Positioned at the top inside of the bin, this sensor measures the distance between the sensor and the surface of the accumulated waste. As the bin fills up, the distance decreases. When the fill level crosses a predefined threshold, the system sends an alert (via the cloud) indicating that the bin is nearing capacity. This feature prevents overflow, enhances cleanliness, and supports efficient waste collection scheduling.



2. NodeMCU (ESP8266 Wi-Fi Microcontroller)

The NodeMCU is a compact and powerful development board based on the ESP8266 chip, known for its built-in Wi-Fi capabilities. In this project, the NodeMCU acts as the central processing unit (CPU) or "brain" of the system.

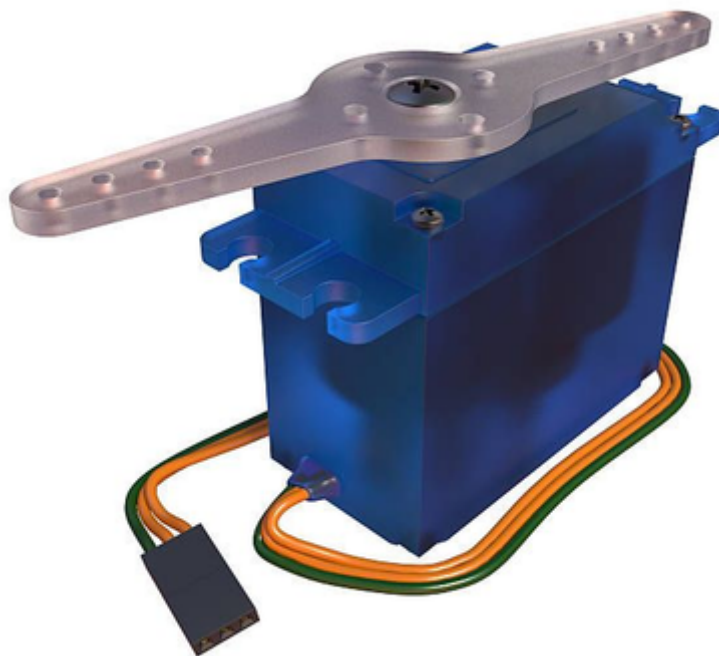
- It interfaces with both ultrasonic sensors and the servo motor.
- It processes incoming data (motion detection, bin level measurement) and makes decisions based on programmed logic.
- It connects to cloud platforms such as Blynk and ThingSpeak to upload real-time data, send alerts, and allow remote monitoring of bin status via a smartphone app or web dashboard.
- This component enables IoT integration, making the system part of the smart city infrastructure.



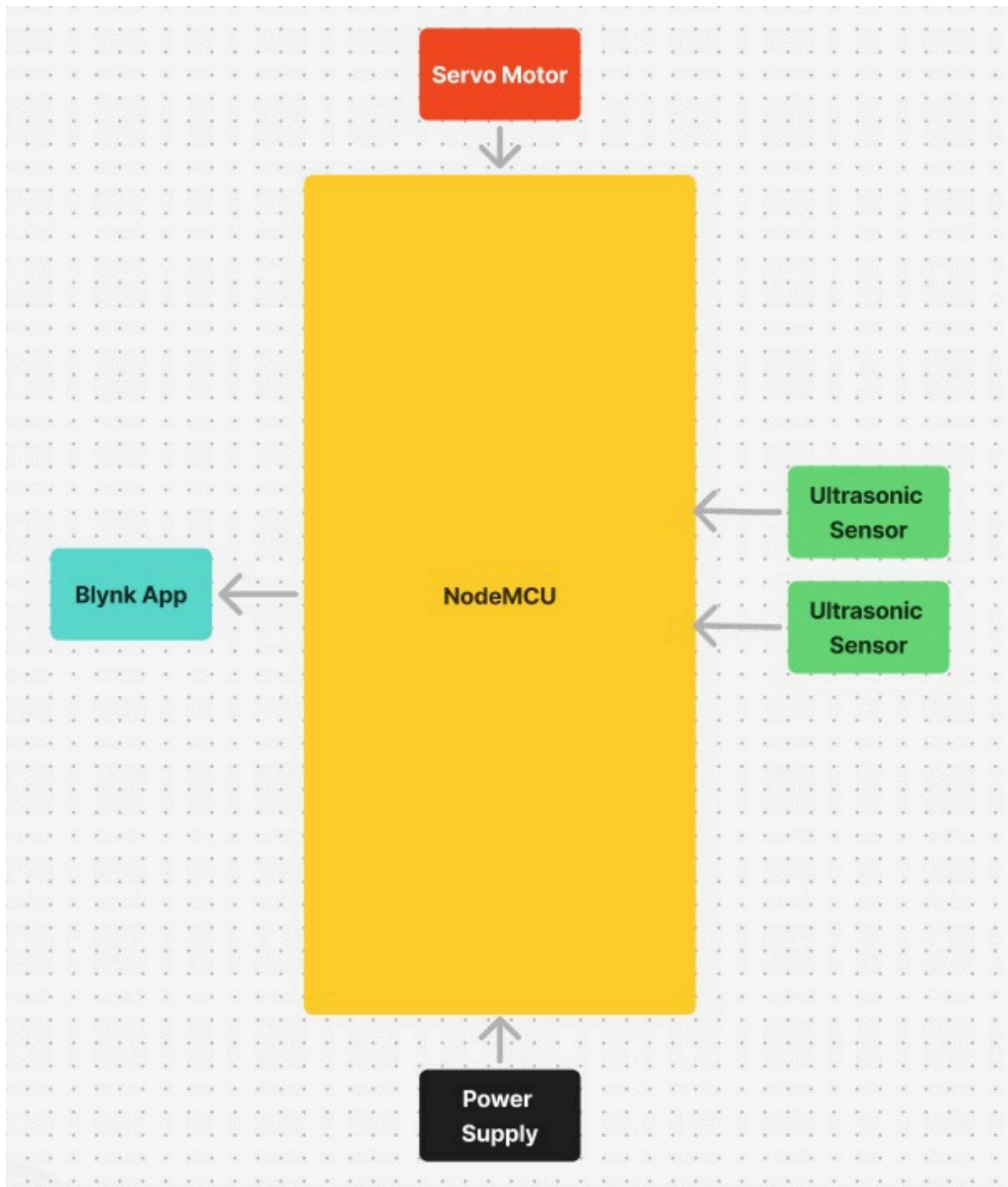
3. Servo Motor

A servo motor is a rotary actuator that allows for precise control of angular position. In this system:

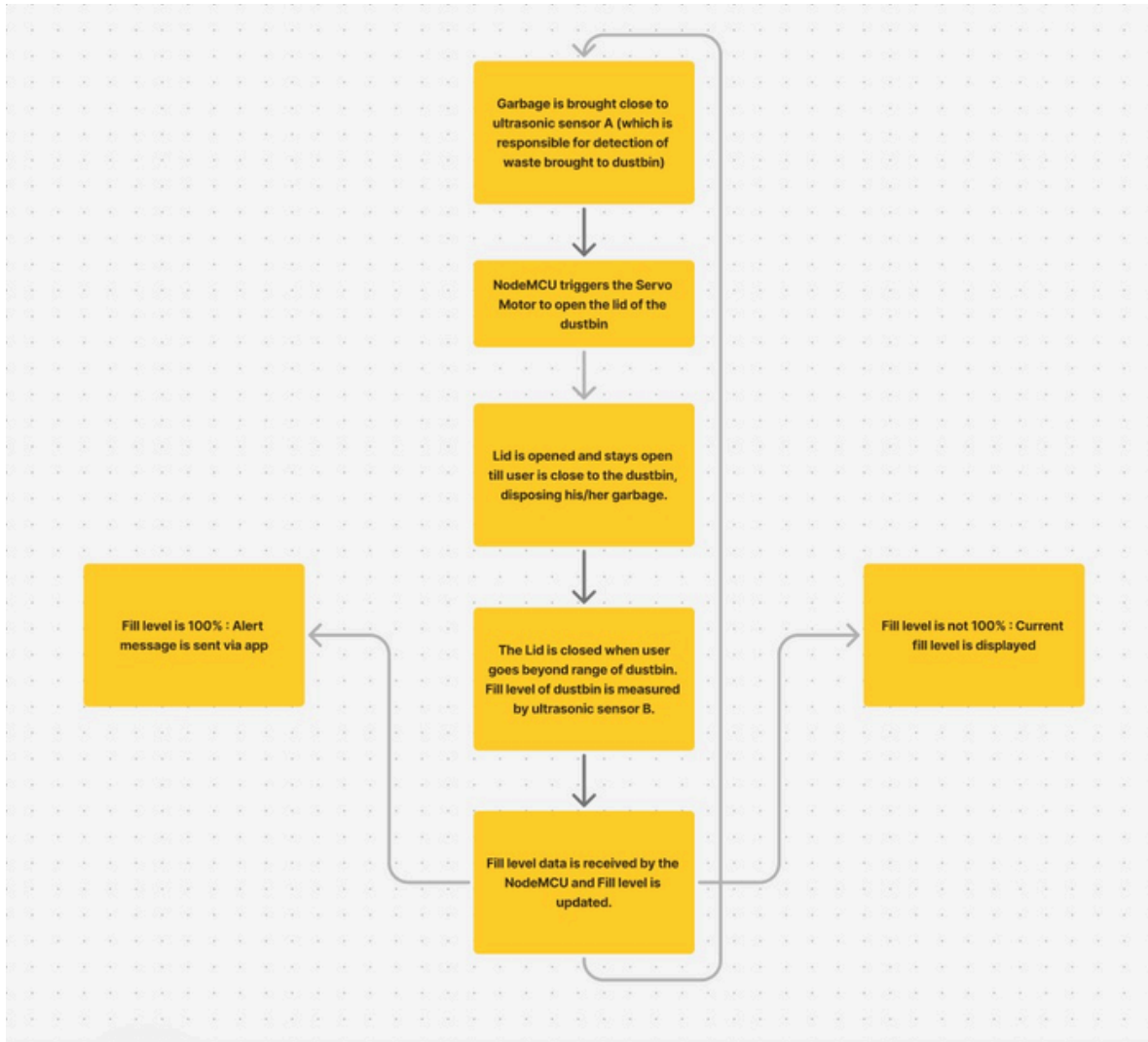
- The servo motor is mechanically connected to the dustbin lid.
- When motion is detected by Sensor 1, the NodeMCU sends a signal to rotate the servo motor to a specified angle (e.g., 90°) to open the lid.
- After a short delay or once the person moves away, the motor returns to its original position to close the lid.
- This automated opening/closing mechanism improves user experience by reducing the need for manual handling, promoting sanitation, and adding an element of smart automation to the system.



BLOCK DIAGRAM :



FLOW OF PROJECT



IMPLEMENTATION

Smart Dustbin – Working Overview

The Smart Dustbin operates using ultrasonic sensors and a servo motor to enable automatic, touch-free waste disposal. It enhances hygiene by detecting objects near the bin and opening the lid automatically.

Object Detection & Lid Operation:

An ultrasonic sensor is placed at the front of the dustbin to detect when a person or object (such as waste) comes close.

If an object is detected within 20 cm, the servo motor rotates to open the lid, allowing waste to be disposed of.

Waste Level Monitoring:

A second ultrasonic sensor inside the bin measures the fill level by calculating the distance between the sensor and the waste surface.

As waste accumulates, the sensor monitors the decreasing distance to determine how full the bin is.

User Alerts & Smart Features:

When the dustbin reaches its maximum capacity, the system can trigger alerts (such as a buzzer, LED, or notification via Wi-Fi).

The system prevents the lid from opening further when the bin is full, ensuring no overflow.

Power & Operation:

The system runs on a NodeMCU (ESP8266) microcontroller, which controls the sensors and servo motor.

The ESP8266 can be further integrated with Wi-Fi to send notifications or updates to a mobile app or cloud-based dashboard for remote monitoring.



The lid remains closed until the user sensor detects an object.



The lid opens when user sensor detects an object.

Results and Discussion

Results:

The Autamed Dustbin prototype was successfully developed and tested using the NodeMCU (ESP8266), ultrasonic sensors, servo motors, and the Blynk IoT platform. The system was evaluated based on parameters like bin fill-level detection, real-time updates, lid automation, and remote monitoring. The following observations were recorded:

Lid Automation:

The ultrasonic sensor mounted at the top of the bin reliably detected approaching objects (such as a user's hand) within a range of 0 to 25 cm. Upon detection, the servo motor opened the lid, and after 5 seconds of inactivity, the lid automatically closed.

The response time was quick (approximately 1 second) and worked consistently in both indoor and semi-outdoor environments.

Fill-Level Monitoring:

A second ultrasonic sensor is installed vertically inside the dustbin to monitor the fill level in real-time. The sensor continuously measures the distance from the lid (sensor location) to the top of the waste pile. Instead of displaying the fill level in percentage terms (e.g., 80% full), the system operates using a threshold-based alert mechanism for simplicity and faster decision-making.

Real-time values were displayed on the Blynk app, along with notifications when the bin reached a full state.

IoT Monitoring with Blynk:

The Blynk mobile application displayed the fill percentage, lid status (Open/Closed), and system status (Active/Idle).

Notifications were successfully triggered when the bin was full or if there was an obstruction inside the bin.

Power and Connectivity:

The system was powered using a portable power bank (5V USB) and tested for up to 8 hours.

Wi-Fi connectivity via NodeMCU was stable with minimal packet loss during operation.

Discussion:

The results demonstrated that the Autamed Dustbin system is a feasible solution for automating waste disposal and monitoring. The use of ultrasonic sensors provided accurate distance readings, essential for both lid automation and fill-level detection.

Advantages Identified:

Hygienic Operation: Touchless interaction reduces the risk of germ transmission.

Efficient Waste Management: Real-time monitoring helps optimize waste collection schedules.

Scalability: The system can be easily deployed across smart homes, offices, or public bins.

Challenges Faced:

Sensor Accuracy Fluctuations: Inconsistencies in sensor readings were occasionally observed due to external light interference or non-uniform waste surfaces.

Wi-Fi Dependency: The system's IoT features are dependent on stable internet access, which may limit deployment in areas with poor connectivity.

Power Constraints: Continuous operation on battery requires energy optimization or solar integration for longer deployments.

Future Enhancements:

Integration with a GSM module for areas without Wi-Fi.

Addition of a weight sensor to complement the ultrasonic sensor for more accurate fill-level estimation.

Implementation of machine learning for smart waste classification and management.

CONCLUSION

The Smart Automatic Dustbin is an innovative solution that enhances hygiene, convenience, and efficiency in waste management. By utilizing sensors and motors, it eliminates the need for manual contact, reducing the risk of contamination and promoting a cleaner environment. The addition of smart features such as waste level detection, IoT connectivity, and odour control further improves its functionality.

With increasing concerns about waste management and sanitation, this smart dustbin can be effectively implemented in homes, offices, hospitals, and public places. It not only simplifies waste disposal but also contributes to sustainable living and smart city development. Future enhancements, including AI-based waste segregation and solar-powered operation, can further improve its impact, making it an essential innovation for modern society.

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